



Solar Inverter Product Overview

Solar inverters, monitoring and support
for PV installations of all sizes

Full range of solar inverters for various applications

Residential, Commercial and Utility-Scale



H2.5 / H3A / H4A /
H5A Flex



M6A / M8A / M10A



RESIDENTIAL



M10A



M15A / M20A / M30A Flex



M50A / M70A / M100A Flex



COMMERCIAL

Photo: PRE Energy GmbH



M125HV Gen2



M250HV



UTILITY-SCALE

MyDeltaSolar Cloud

Monitoring for your PV installation

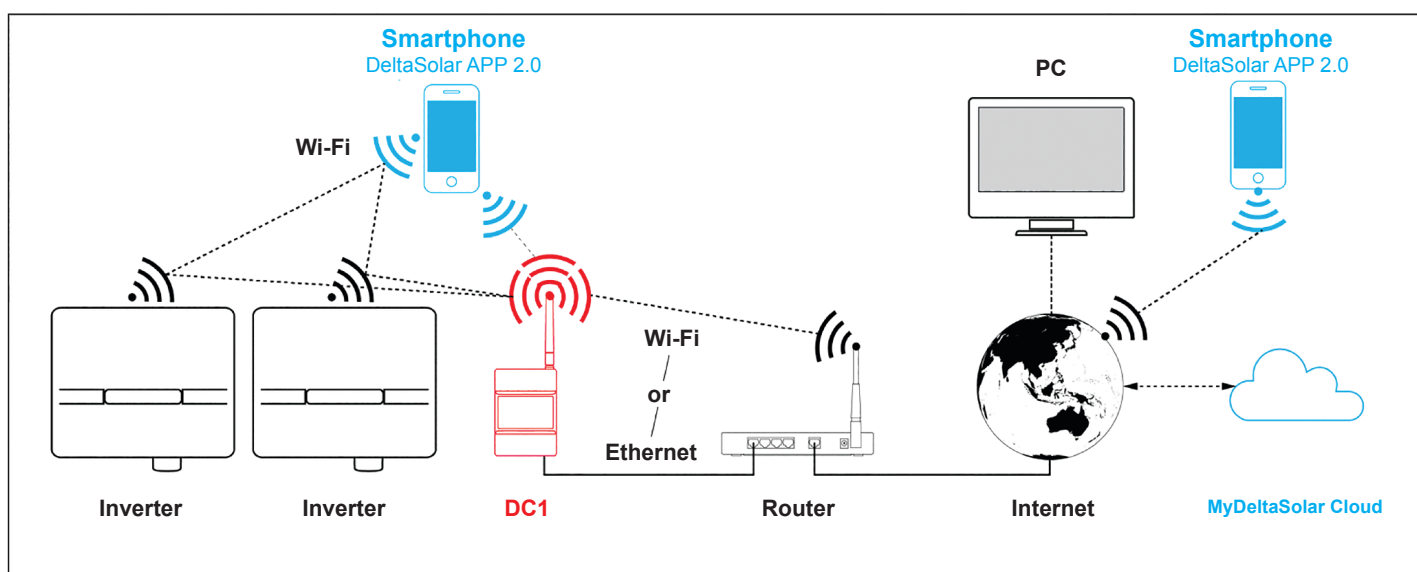
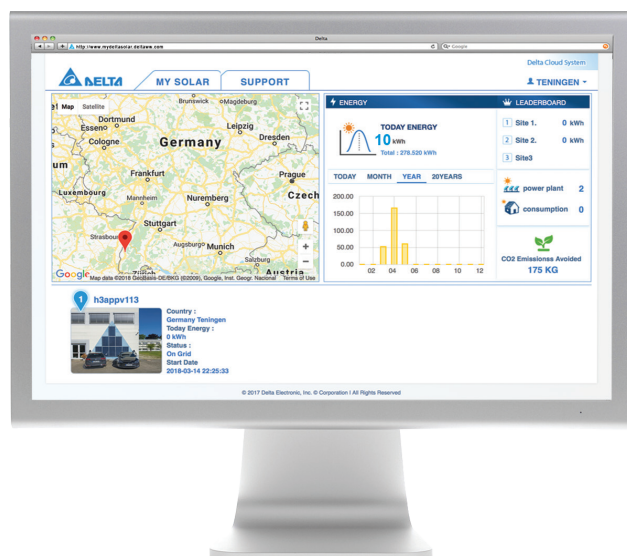
MyDeltaSolar Cloud is a cloud-based monitoring system that allows the solar inverter owner to register their PV plant online to view the status anytime and anywhere. This ensures reliable operation and a maximum return on your investment in your solar system.

The application can be used with Delta inverters featuring Wi-Fi communication such as the Flex-series single-phase inverters and RS485 inverters via our DC1 Data Collector.

The user-friendly internet application provides a quick overview of key parameters, such as yield and system status, making it perfect for residential or small commercial PV systems.

The performance data is gathered from the inverters, sent over the internet, and presented on your computer or web-enabled device with easy to read graphs.

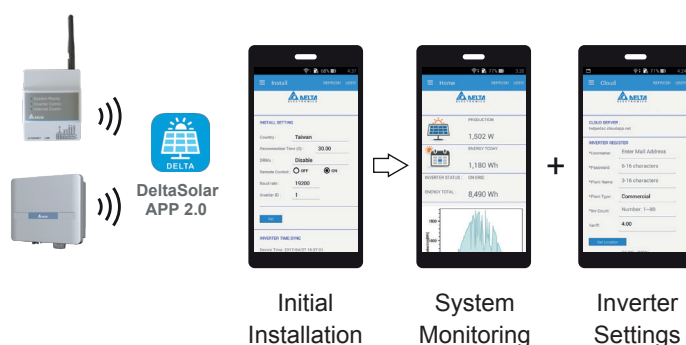
Automated email alerts can be configured to provide immediate notification of any issues, avoiding downtime on your solar investment.



DeltaSolar App 2.0

App for monitoring your PV system via Wi-Fi or RS485 connection with the DC1 Data Collector

The DeltaSolar App uses the integrated Wi-Fi communication of the single-phase Flex series to connect to the inverter for commissioning and monitoring your PV system. The app can be used in conjunction with the DC1 Data Collector both with our single-phase Flex inverters via Wi-Fi or also via the RS485 interface of the 3ph. inverters and is available for Android and iOS mobile devices.



The Delta Advantage

More than just great products



Strong and reliable company with local presence

- Experienced contact partners in your area and in your language
- Knowledge of the local grid access conditions



Highly reliable products with extensive testing and continuous improvement

- Extensively tested products in extreme conditions
- Robust electronics and die-cast aluminum chassis
- IP65 protection class for outdoor installation



Extensive support for sales partners, installers, and plant operators

- Delta Solar Team offers inverter training courses which can be run on your premises upon request



Delta offers reliable, flexible and competent support

- Service with own employees
- Qualified technicians with years of PV experience
- Direct contact to technical support (Mon. to Fri., 8 am to 5 pm):
Intl.: +49 (0)7641 455 547
support@solar-inverter.com

2.5 kVA - 5 kVA for residential PV systems

H2.5 Flex	H3A / H4A Flex	H5A Flex
 ✓ Wi-Fi comm. ✓ Ultra Low Startup Voltage	 ✓ Wi-Fi comm. ✓ Ultra Low Startup Voltage	 ✓ Wi-Fi comm. ✓ Ultra Low Startup Voltage

	H2.5 Flex	H3A Flex	H4A Flex	H5A Flex
Input (DC)				
Nominal power (W)	2600	3190	4320	5280
Voltage range (V)	30 ... 500	30 ... 600		
MPP voltage range (full power) (V)	240 ... 470	180 ... 500	240 ... 500 ¹⁾	
Max. number of MPP Trackers	1	2 ²⁾		
Output (AC)				
Number of phases	1			
Nominal apparent power (VA) ³⁾	2500	3000	4000	5000 ⁴⁾
Voltage range (V) ⁵⁾	230 (-20% / +22 %), 1 phase (L, N, PE)			
Nominal frequency (Hz) ⁵⁾	50 / 60 ± 5			
Power factor adjustable	0.8 capacitive ... 0.8 inductive			
Additional Information				
Max. efficiency (%)	97.5			98.3
Efficiency EU (%)	96.8			98.0
Operating temperature (°C)	-25 ... +60 (full power up to +40 °C)			
Protection degree	IP65			

¹⁾ When operated with maximum current

²⁾ Parallel inputs: 1 MPP Tracker, Separate inputs: 2 MPP Trackers, input load: symmetrical and asymmetrical (for H3A Flex / H4A Flex: 0/100 %, for H5A Flex: 4/96 %)

³⁾ Cos Phi = 1 (VA = W)

⁴⁾ 4.6 kW for Germany

⁵⁾ AC voltage and frequency range will be programmed according to the individual country requirements

Delta Solar Inverters

Exceptional performance and wide range of functions

Maximum profitability



Delta solar inverters achieve a peak efficiency of up to 99.2 % and offer one of the largest operating temperature ranges available. Full output up to 50 °C ambient temperature to maximise energy efficiency and ensure maximum profitability.

Long lifetime



Delta inverters have a very long lifetime thanks to the durable aluminum housing and high quality robust electronics that are employed in their design. Continuous product improvement is ensured due to comprehensive testing during product development and in production.

Special characteristics



One of its special features is the anti-PID (Potential Induced Degradation) function. This function protects solar modules from power losses due to PID, by allowing reactive power to be delivered at night.

It is possible to replace the integrated type 2 surge protector with one of type 1+2. Monitoring allows one to read the values for the display of the V-I curve for each individual string.

Aluminium housing



The aluminium housing of Delta solar inverters ensure long lasting protection against moisture and corrosion. The dust-proof, completely shock-proof housing protects the electronics of the solar inverter from water spray from all directions. This means that it can be installed in protected outdoor areas, damp basements and areas with a high level of dust (ie. agriculture).

6 kVA - 10 kVA for residential PV systems

M6A	M8A	M10A
 ✓ Wi-Fi and RS-485 comm.	 ✓ Wi-Fi and RS-485 comm.	 ✓ Wi-Fi and RS-485 comm.

	M6A	M8A	M10A
Input (DC)			
Nominal power (W)	6600	8800	11000
Voltage range (V)	200 ... 1000		
MPP voltage range (full power) (V) ¹⁾	315 ... 850	415 ... 800	
Max. number of MPP Trackers	2 ²⁾		
Output (AC)			
Number of phases	3		
Nominal apparent power (VA) ³⁾	6000	8000	10000
Voltage range (V) ⁴⁾	230 ± 20 % / 400 V _{AC} ± 20 %; 3 phase + PE or 3 phase + N + PE		
Nominal frequency (Hz) ⁴⁾	50 / 60 ± 5		
Power factor adjustable	0.8 capacitive ... 0.8 inductive		
Additional Information			
Max. efficiency (%)	98.3	98.3	98.3
Efficiency EU (%)	97.6	97.9	98.0
Operating temperature (°C)	-25 ... +60 (full power up to +40 °C)		
Protection degree	IP65		

¹⁾ When operated with balanced DC inputs (50/50 %)

²⁾ Parallel inputs: 1 MPP Tracker, Separate inputs: 2 MPP Trackers, input load: symmetrical and asymmetrical (for M6A, M8A, M10A: 40/60 %)

³⁾ Cos Phi = 1 (VA = W)

⁴⁾ AC voltage and frequency range will be programmed according to the individual country requirements

Monitoring of your PV plant

DC1 Data Collector

- Collect all relevant data from your Delta inverters
- Manage multiple PV plants in the MyDeltaSolar cloud
- Connect to your Delta inverters via RS485 or Wi-Fi
- Compatible with third-party monitoring solutions
- Digital inputs, potential-free contacts, dynamic power limitation



GENERAL	DC1
Supply voltage (RS485 plug)	9 ... 25 VDC / 5 W (from communication card of inverter or Delta DIN rail Power Supply DRC-12V10W1AZ or equivalent)
Supply voltage (Micro-USB)	5 V _{DC} / 5 W
Protection degree	IP20
Relative humidity	30 ... 85%
Operating temperature range	-25 ... +55 °C
Storage temperature range	-30 ... +70 °C
Maximum operating altitude	2000 m
Max. number of inverters	RS485: 30; Wi-Fi: 9
Connectors	RS485, Wi-Fi, Micro-SD, Extension port, USB 2.0

Wi-Fi	DC1
Network standards	IEEE 802.11b IEEE 802.11g IEEE 802.11n
Data rates	IEEE 802.11b: 6.5 ... 150 Mbps IEEE 802.11g: 6 ... 54 Mbps IEEE 802.11n: 1 ... 1 Mbps
Modulation	OFDM (BPSK), QPSK, 16-QAM, 64-QAM; 802.11b (CCK / DSSS)
Bandwidth	20 MHz / 40 MHz

15 kVA - 100 kVA for commercial PV installations

M15A / M20A Flex	M30A Flex	M50A / M70A / M100A Flex
 NEW! ✓ Low noise (max. 31.6 dBA) ✓ Wireless Comm.	 NEW! ✓ 3 MPP Trackers ✓ Wireless Comm.	 NEW! ✓ 6 MPP Trackers (M50A/M70A) ✓ 8 MPP Trackers (M100A) ✓ Wireless Comm. (optional)

	M15A Flex	M20A Flex	M30A Flex	M50A Flex	M70A Flex	M100A Flex
Input (DC)						
Nominal power (W)	16500	22000	33000	55000	77000	110000
Voltage range (V)	200 ... 1000					
MPP voltage range (full power) (V)	325 ... 900	460 ... 900	480 ... 900	390 ... 900	460 ... 900	520 ... 840
Max. number of MPP Trackers	2	2	3	6		8
Output (AC)						
Number of phases	3					
Nominal apparent power (VA) ¹⁾	15000	20000	30000	50000	77000	100000
Voltage range (V) ²⁾	230/400 V -20 % / +30 %; 3 phase + PE (Δ) or 3 phase + N + PE (Y)					
Nominal frequency (Hz) ²⁾	50 / 60 ± 5					
Power factor adjustable	0.8 capacitive ... 0.8 inductive					
Additional Information						
Max. efficiency (%)	98.3	98.5	98.6	98.7	98.8	98.7
Efficiency EU (%)	97.9	98.1	98.2	98.3	98.4	98.5
Operating temperature (°C)	-25 ... +60 ³⁾					
Protection degree	IP66					

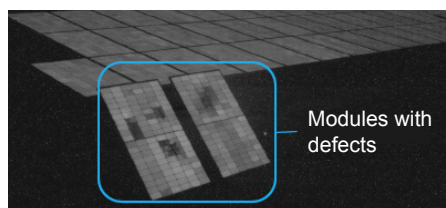
¹⁾ Cos Phi = 1 (VA = W)

²⁾ AC voltage and frequency range will be programmed according to the individual country requirements

³⁾ Full nominal power without derating up to +50°C for M15A Flex, +50°C for M20A Flex, +40°C for M30A Flex, +50°C for M50A Flex, +50°C for M70A Flex with nominal voltages (AC and DC) and Cos Phi =1

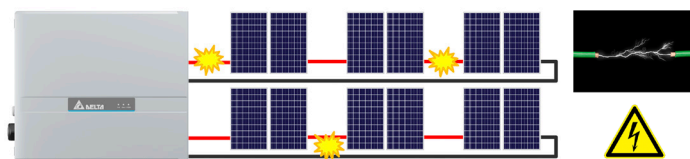
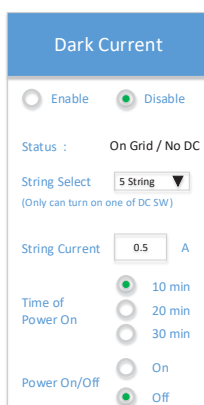
Special technical functions

Pro-EL function and arc fault detection



Unique Pro-EL electroluminescence function

- Available for M125HV and M250HV
- Early detection of defective cells and modules (e.g. due to microcracks)
- Based on the functionality of an LED
- Current flows into the solar cells, causing light to be emitted (1150 nm)
- Requires an infrared capable CCD camera & smartphone with DeltaSolar APP 2.0. Controllable via Delta Service Software.



Integrated arc fault detection in 15 - 250 kVA solar inverters

- Available for M15A Flex / M20A Flex / M30A Flex / M50A Flex / M70A Flex / M100A Flex / M125HV and M250HV
- Delta Electronics focuses primarily on the detection of serial arc faults, as this is the most likely fault to occur.
- The arc fault detection exhibits an interruption time well below 1s (UL 1699B section 29).

125 kVA - 250 kVA for utility-scale PV installations

M125HV Gen2	M250HV
 NEW! ✓ 140 kVA ✓ 1500 Vdc	 NEW! ✓ 250 kVA ✓ 1500 Vdc ✓ 12 MPP Trackers

	M125HV Gen2	M250HV
Input (DC)		
Nominal power (W)	125000	250000
Voltage range (V)	860 ... 1500	500 ... 1500
MPP voltage range (full power) (V)	860 ... 1450	820 ... 1350
Max. number of MPP Trackers	1	12
Output (AC)		
Number of phases	3	
Nominal apparent power (VA) ¹⁾	140000 VA / 125000 W Active Power	250000
Voltage range (V) ²⁾	600 V -36% / +15%; 3 phase + PE (Δ)	640 to 920 VAC; 3 phase + PE (Δ)
Nominal frequency (Hz) ²⁾	50 / 60 ± 5	
Power factor adjustable	0.8 capacitive ... 0.8 inductive	
Additional Information		
Max. efficiency (%)	99.1	99.0
Efficiency EU (%)	98.7	98.8
Operating temperature (°C)	-25 ... +60 ³⁾	
Protection degree	IP66	

¹⁾ Cos Phi = 1 (VA = W)

²⁾ AC voltage and frequency range will be programmed according to the individual country requirements

³⁾ Full nominal power without derating up to +50°C for M125HV, +50°C for M250HV with nominal voltages (AC and DC) and Cos Phi=1

Delta Service & Support

Reliable, Flexible and Competent

Good service is very important. That is why Delta offers reliable, flexible and competent support.

Customer satisfaction and short response times are not just words for Delta. Delta takes care of partners and customers over the complete lifetime of the products.

Our highly motivated solar team supports Delta customers via a dedicated Solar Support hotline, which is always there for you when you need help.

If necessary, our service teams will also support you on site. Replacement equipment and components are available within two business days.

Do you need detailed information about the solar inverter? No problem! Our Delta Solar team offers solar inverter seminars, which can be held either at our premises or at your site upon request.



Sales Contact

United Kingdom

sales.uk@solar-inverter.com
0800 051 4280 (Free Call)

Other European countries

sales.europe@solar-inverter.com
+49 7641 455 547

